

Mobile and Embedded Computing

Laboratory 1. Orientation, Set-up & Project

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- [Teams](#)

45+ Flutter apps launched

400k installs across all apps

Appeared in, speaker at & partners



Laboratory Grading (4.5p total)

3p Laboratory + Assignments (individual)

1.5p Lab test

Project Grading (1.5p total)

+ 1.5p Project (teams of maximum 3)

Project details & deliverables

- Project development MUST be **tracked on GitHub**
- The application MUST be **written in Flutter**
- The way of working will also be marked (Pull Requests, Branches, Code reviews, GitHub Issues, etc.)
- Individual participation will be measured based on Commits, Lines of Code, and the quality of the code written. **Each team member** should know **in detail what he has implemented** and have at least a brief awareness of the other parts of the project
- You will propose your own project, and we'll discuss and approve it together in the next lab
- All your presentation materials (pptx and also convert it to .pdf) should be uploaded in GitHub in a folder named presentation
- Teams of max 3 students
- You will present from your own laptop, at the projector

- Think of a project idea and form teams in the given Google Sheet
- After forming the teams and writing a description in the doc we will have a discussion to validate its complexity
- Your project should have a medium-hard complexity

Your project must at least have the following:

- State management using BLoC
- Authentication (Firebase Auth / Keycloak / anything that is 'real' auth)
- Data persistence (any kind of database) and offline first capabilities
- When logging in on a different device, user's data should be loaded/transferred automatically
- Realtime capabilities using Websockets

- The readme MUST contain the following sections (in this order):
 - **Team composition**, 1 person / row (Name, Surname, Group)
 - **Project description** (brief) of the implemented app (what it does, what 'screens' it has, etc.)
 - **Link to the public GitHub repository** (or if you chose to make the repo private, make sure to invite me as a collaborator, my username is: rusudinu)
 - **Other information** (anything else you would like to mention)
- The final presentation will be a live demo accompanied by slides and must not take more than 15 minutes / team. Questions will be asked based on the code; you should NOT copy entire projects from the internet, and you should not vibe code stuff that you can't explain later or do not already understand.

Environment set-up



Tasks

- Set-up Flutter and your development environment
 - Install Flutter by following the documentation (<https://docs.flutter.dev/install>) (Click on Install Manually in the Install Flutter section)
 - Android Studio / Xcode (<https://developer.android.com/studio>)
 - IntelliJ / VSCode to write the code (with the Flutter and Dart plugins)
 - Configure Android / iOS emulators
- Validate development environment by creating and running the default Flutter project, it should open in your emulator